

HU Collection

TMUA Mock Set I Paper 1

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	D	D	C	A	E	A	D	F	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	A	B	B	C	E	C	A	A	D	B

Each solution gives the intended route and names the trap the question was built around. Questions are ordered by increasing difficulty.

Question 1. Answer B.

Differentiate implicitly: $2x + y + (x + 2y)\frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{2x+y}{x+2y}$. Setting this equal to 1 gives $2x + y = -x - 2y$, that is $y = -x$. On the curve, $x^2 - x^2 + x^2 = 3$, so the points are $(\sqrt{3}, -\sqrt{3})$ and $(-\sqrt{3}, \sqrt{3})$, at distance $\sqrt{12 + 12} = 2\sqrt{6}$. As a check on the geometry: the curve is symmetric in the line $y = x$, so the tangents parallel to that line occur where the curve crosses the perpendicular line $y = -x$.

Question 2. Answer D.

The geometric condition is $(a + 4d)^2 = (a + d)(a + 10d)$, which simplifies to $6d^2 = 3ad$, so $a = 2d$ (the common ratio is then $\frac{6d}{3d} = 2$). The n th term is $a + (n - 1)d = (n + 1)d$, and the sum of ten terms is $5(2a + 9d) = 65d$. Setting $(n + 1)d = 65d$ gives $n = 64$. The trap is to see $65d$ and answer “the 65th”: the off-by-one comes from $a = 2d$ rather than $a = d$.

Question 3. Answer D.

The left side is the sum of the distances from x to a and to $2a$. Its minimum value is $|a|$, the distance between those two points, attained for every x between them, and it increases linearly outside. So there are no solutions if $a^2 < |a|$, that is $0 < |a| < 1$. There are infinitely many, a whole interval, if $a^2 = |a|$ with $a \neq 0$, that is $a = \pm 1$. There are exactly two, one on each side, if $a^2 > |a|$, that is $|a| > 1$. At $a = 0$ the equation is $2|x| = 0$, with the single solution $x = 0$.

Question 4. Answer C.

Stationary points satisfy $4x^3 - 16x + a = 0$, that is $a = 16x - 4x^3 = h(x)$. The graph of h has a local minimum at $x = -\frac{2}{\sqrt{3}}$ and a local maximum at $x = \frac{2}{\sqrt{3}}$, of value $\frac{32}{\sqrt{3}} - \frac{32}{3\sqrt{3}} = \frac{64}{3\sqrt{3}} \approx 12.3$. Three distinct stationary points need $|a| < \frac{64}{3\sqrt{3}}$, so the integers $-12, \dots, 12$: 25 values. At the boundary two stationary points merge, but $\frac{64}{3\sqrt{3}}$ is not an integer, so no integer is lost or gained there.

Question 5. Answer A.

Let $g(x) = x^2 e^{-x}$. Then $g'(x) = x(2 - x)e^{-x}$: a minimum $g(0) = 0$ and a maximum $g(2) = \frac{4}{e^2}$. As $x \rightarrow -\infty$, $g \rightarrow +\infty$, and as $x \rightarrow +\infty$, $g \rightarrow 0$ from above. So a level $0 < k < \frac{4}{e^2}$ is met once for

$x < 0$, once on $(0, 2)$ and once for $x > 2$, where g decays towards 0 but never reaches it. That gives three solutions. At $k = \frac{4}{e^2}$ there are two; at $k = 0$ one; for $k > \frac{4}{e^2}$ only the branch $x < 0$ remains. The third root, on the long tail, is the one people miss.

Question 6. Answer E.

$f(0) = 0$ gives $c = 0$, and $f(1) = 1$ gives $b = -a$. Then $f'(x) = 3x^2 + 2ax - a \geq 0$ for all x exactly when the discriminant $4a^2 + 12a \leq 0$, that is $-3 \leq a \leq 0$. And $f(2) = 8 + 4a + 2b = 8 + 2a$, which ranges over $[2, 8]$. Both ends are attained: $a = 0$ gives x^3 , and $a = -3$ gives $x^3 - 3x^2 + 3x = (x - 1)^3 + 1$, whose stationary point of inflection still counts as increasing.

Question 7. Answer A.

Three real roots need $a > 0$, with turning points at $x = \pm\sqrt{a}$. The local maximum value is $f(-\sqrt{a}) = 2 + 2a\sqrt{a} > 0$ always, and the local minimum value is $f(\sqrt{a}) = 2 - 2a\sqrt{a}$, which is negative exactly when $a > 1$. The smallest root exceeds -3 exactly when $f(-3) < 0$ and -3 lies to the left of the local maximum ($\sqrt{a} < 3$). $f(-3) = 9a - 25 < 0$ gives $a < \frac{25}{9}$, which also ensures $a < 9$. So $1 < a < \frac{25}{9}$.

Question 8. Answer D.

Split triangle ABC along AD into two triangles with angles 60° at A , and add their areas: $\frac{1}{2}xy \sin 120^\circ = \frac{1}{2} \cdot 2x \sin 60^\circ + \frac{1}{2} \cdot 2y \sin 60^\circ$. The sines cancel, leaving $xy = 2(x + y) = 24$. This is the pleasant fact that a 120° bisector has length $\frac{xy}{x+y}$. Then the cosine rule gives $BC^2 = x^2 + y^2 + xy = (x + y)^2 - xy = 144 - 24 = 120$, so $BC = 2\sqrt{30}$. Neither x nor y needs to be found.

Question 9. Answer F.

For $a \geq 3$ the integrand is $ax - x^2$ throughout, and $I(a) = \frac{9}{2}a - 9$ is increasing, so it is at least $\frac{9}{2}$. For $0 < a < 3$, split at $x = a$: $\int_0^a (ax - x^2) dx = \frac{a^3}{6}$ and $\int_a^3 (x^2 - ax) dx = 9 - \frac{9}{2}a + \frac{a^3}{6}$, so $I(a) = \frac{a^3}{3} - \frac{9}{2}a + 9$. This has $I'(a) = a^2 - \frac{9}{2} = 0$ at $a = \frac{3}{\sqrt{2}}$, giving $I = 9 - \frac{9}{\sqrt{2}} = \frac{18-9\sqrt{2}}{2} \approx 2.64$, less than $\frac{9}{2}$. Option A is the value at $a = 3$.

Question 10. Answer C.

The tangent is $y = 2tx - t^2$. It meets $y = x^2 - k$ where $x^2 - 2tx + t^2 - k = 0$, that is $(x - t)^2 = k$, so $x = t \pm \sqrt{k}$: the chord always has horizontal width $2\sqrt{k}$, whatever t is. The region between a line and the parabola $y = x^2 - k$ over an interval of width w has area $\frac{w^3}{6}$ (the quadratic difference is $(x - x_1)(x_2 - x)$). So the area is $\frac{(2\sqrt{k})^3}{6} = \frac{4}{3}k^{3/2} = 36$, giving $k^{3/2} = 27$ and $k = 9$. Seeing why the area cannot depend on t is the heart of the question.

Question 11. Answer A.

Write $3\sin x + 4\cos x = 5\sin(x + \alpha)$, where $\tan \alpha = \frac{4}{3}$ and $0 < \alpha < \frac{\pi}{2}$, so $\sin \alpha = \frac{4}{5}$. As x runs from 0 to π , $x + \alpha$ runs from α to $\pi + \alpha$, so the expression starts at 4, rises to 5 at $x = \frac{\pi}{2} - \alpha$, and then falls steadily to -4 at $x = \pi$. A level k is met twice when $4 \leq k < 5$ (once on the rise, which includes $x = 0$ when $k = 4$, and once on the fall), once at $k = 5$, and once for $-4 \leq k < 4$, on the fall only. So the answer is $-4 \leq k < 4$ or $k = 5$. The endpoint $k = 4$ is the trap: $x = 0$ is a genuine solution there.

Question 12. Answer B.

Counting residue patterns works but is long. Instead, consider the map that adds 1 to each chosen number, sending $3m$ back to 1. It permutes the four-element subsets, and it changes each chosen number by an amount that is 1 modulo 3 (including the wrap-around $3m \rightarrow 1$, a change of $1 - 3m$). So it changes the sum by $4 \equiv 1 \pmod{3}$. It therefore maps the subsets with sum $\equiv 0$ one-to-one onto those with sum $\equiv 1$, and those onto sum $\equiv 2$. All three classes are equally large, and the probability is exactly $\frac{1}{3}$ for every m . For three numbers the same shift changes the sum by $3 \equiv 0$, which is why that case gives $\frac{68}{203}$ for $m = 10$ rather than $\frac{1}{3}$.

Question 13. Answer B.

Two parameter values give the same $x = t^2 - 1$ only if they are t and $-t$, and since $y(-t) = -y(t)$, the points coincide only when $y = t(t^2 - 1) = 0$ with $t \neq 0$, so $t = \pm 1$ and the crossing point is $(0, 0)$. The loop is traced for $-1 \leq t \leq 1$, and its area is

$$\left| \int_{-1}^1 y \frac{dx}{dt} dt \right| = \left| \int_{-1}^1 (t^3 - t)(2t) dt \right| = 2 \left| \frac{2}{5} - \frac{2}{3} \right| = \frac{8}{15}.$$

At the crossing the tangent gradients are $\frac{3t^2-1}{2t} = \pm 1$, so the two branches cross at right angles, a pleasant check on any sketch.

Question 14. Answer C.

The overlap is the original square minus four congruent right-angled triangles at its corners. Each has legs p and $q = p \tan \theta$, and its hypotenuse lies along a side of the rotated square, so along each side of the original square $p + q + \sqrt{p^2 + q^2} = 1$. This gives $p = \frac{\cos \theta}{1 + \sin \theta + \cos \theta}$ and $q = \frac{\sin \theta}{1 + \sin \theta + \cos \theta}$. With $s = \sin \theta + \cos \theta$, so that $2 \sin \theta \cos \theta = s^2 - 1$, the overlap is $1 - 2pq = 1 - \frac{s^2 - 1}{(1 + s)^2} = \frac{2}{1 + s}$. Setting this to $\frac{4}{5}$ gives $s = \frac{3}{2}$, so $\sin \theta \cos \theta = \frac{5}{8}$ and $\tan \theta + \frac{1}{\tan \theta} = \frac{1}{\sin \theta \cos \theta} = \frac{8}{5}$.

Question 15. Answer E.

Let $g(x) = x^2 + 2ax + b$. The conditions are $a^2 > b$ (distinct real roots), the vertex $-a$ in $(-1, 1)$ (automatic), $g(1) = 1 + 2a + b > 0$ and $g(-1) = 1 - 2a + b > 0$. So $2|a| - 1 < b < a^2$, and this interval is never empty, since $a^2 - 2|a| + 1 = (|a| - 1)^2 \geq 0$. The favourable area is $\int_{-1}^1 (|a| - 1)^2 da = 2 \int_0^1 (1 - a)^2 da = \frac{2}{3}$, out of a total area of 4. The probability is $\frac{1}{6}$. Option D forgets to divide by the area of the square.

Question 16. Answer C.

Divide $f(x) = x^3 + ax^2 + bx$ by $f'(x) = 3x^2 + 2ax + b$:

$$f(x) = \left(\frac{x}{3} + \frac{a}{9} \right) f'(x) + \left(\frac{2b}{3} - \frac{2a^2}{9} \right) x - \frac{ab}{9}.$$

At each stationary point $f' = 0$, so both lie on the line $y = \left(\frac{2b}{3} - \frac{2a^2}{9} \right) x - \frac{ab}{9}$, which is therefore the line through them. It passes through the origin exactly when $ab = 0$. (The condition for two stationary points is $a^2 > 3b$, so $a = 0$ requires $b < 0$, while $b = 0$ requires only $a \neq 0$.) Finding the two stationary points explicitly works, but only after heavy surd algebra.

Question 17. Answer A.

No direct substitution works. Let I be the integral and J the same integral with $\cos^3 x$ on top. The substitution $x \mapsto \frac{\pi}{2} - x$ swaps $\sin$ and $\cos$, so $I = J$. Adding, and using $s^3 + c^3 = (s+c)(s^2 - sc + c^2)$:

$$I + J = \int_0^{\pi/2} (1 - \sin x \cos x) dx = \frac{\pi}{2} - \frac{1}{2}.$$

So $I = \frac{\pi-1}{4}$. Option B comes from forgetting the $-sc$ term.

Question 18. Answer A.

$\sin nx = \sin x$ exactly when $nx = x + 2k\pi$ or $nx = \pi - x + 2k\pi$. The first family gives $x = \frac{2k\pi}{n-1}$, which is $n-1$ values in $[0, 2\pi)$; the second gives $x = \frac{(2j+1)\pi}{n+1}$, which is $n+1$ values. That makes $2n$ before removing overlaps. A common value needs $2k(n+1) = (2j+1)(n-1)$. If n is even, the right side is odd and the left even, so there is no overlap. If $n = 4m+3$, the left side is a multiple of 4 and the right side $(2j+1)(4m+2)$ is only a multiple of 2, so again none. If $n = 4m+1$, the values $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ lie in both families, so $N = 2n - 2$. Checking $n = 3$ gives 6 and $n = 5$ gives 8, which rules out C.

Question 19. Answer D.

Expanding, the integral is $J = \frac{1}{5} + \frac{a^2}{3} + b^2 - \frac{a}{2} - \frac{2b}{3} + ab$, a quadratic in two variables. For fixed a , the best b makes $\frac{\partial J}{\partial b} = 2b - \frac{2}{3} + a = 0$, so $b = \frac{1}{3} - \frac{a}{2}$. Substituting leaves a quadratic in a alone, minimised at $a = 1$, $b = -\frac{1}{6}$. Then $J = \frac{1}{5} + \frac{1}{3} + \frac{1}{36} - \frac{1}{2} + \frac{1}{9} - \frac{1}{6} = \frac{36+60+5-90+20-30}{180} = \frac{1}{180}$. The line $y = x - \frac{1}{6}$ is the best straight-line approximation to x^2 on $[0, 1]$ in this sense, and it is not the tangent or the chord, which give larger values.

Question 20. Answer B.

By symmetry, work in the triangle below the centre with vertices $(0, 0)$, $(1, 0)$ and $(\frac{1}{2}, \frac{1}{2})$, where the nearest side is $y = 0$. The condition $y < \sqrt{(x - \frac{1}{2})^2 + (y - \frac{1}{2})^2}$ simplifies to $y < (x - \frac{1}{2})^2 + \frac{1}{4}$, a parabola. It lies above the edge $y = x$ of the triangle until $x = c = 1 - \frac{1}{\sqrt{2}}$. So the favourable area in the left half of the triangle is $\int_0^c x dx + \int_c^{1/2} \left((x - \frac{1}{2})^2 + \frac{1}{4} \right) dx$. Evaluating and multiplying by 8, for two halves times four triangles, gives $\frac{8-4\sqrt{2}}{3} \approx 0.781$. Rough reasoning often suggests $\frac{3}{4}$, which is close but not correct.